Productivity Monitoring System Definition Document

System Overview and Purpose

System Overview

The Productivity Monitoring System is an integrated management system that supports productivity improvement by tracking and analyzing the performance of manufacturing companies' production lines in real-time. This system collects and analyzes production data by company and line in real-time, providing key information necessary for decision-making.

System Purpose

- Real-time monitoring of productivity by company and line for immediate status awareness
- Performance measurement through comparison of target UPH (Units Per Hour) vs actual UPH
- Provision of visualization and analytical information for data-driven productivity improvement
- Support for continuous improvement activities through line-specific achievement rate target management
- Comprehensive dashboard provision for production-related decision support

Key Functions and Features

Real-time Monitoring Function

- Real-time tracking of production volume and UPH by production line
- Real-time calculation and display of achievement rates vs targets
- Detection and notification of production anomalies
- Support for simultaneous monitoring of multiple companies and lines

Data Analysis and Visualization

- Production trend analysis through various charts and graphs
- Line-specific performance comparison analysis
- Defect rate analysis and quality management support
- Work time efficiency analysis

Target Management System

- Setting and management of target UPH by line
- Performance evaluation based on achievement rates
- Gap analysis between targets and actual performance
- Setting and tracking of improvement targets

Data Structure and Integration Plan

Data Sources

This system utilizes view tables from the DATA mart as the primary data source, collecting data through Cursor AI. This ensures accurate and reliable production data acquisition in real-time.

Core Data Columns

Column Name	Data Type	Description
Production Date	DATE	Date when production was performed
Line Name	VARCHAR	Production line identifier
Product Group	VARCHAR	Category of produced products
Production Quantity	INTEGER	Number of products produced in the time period
Defect Quantity	INTEGER	Number of defective products during production
Work Hours	DECIMAL	Actual time spent on work (in hours)
Target UPH	DECIMAL	Target production volume per hour set for each line
Actual UPH	DECIMAL	Actual production volume achieved per hour
Company	VARCHAR	Name of company responsible for production

Data Integration Method

- Real-time integration with DATA mart view tables through Cursor AI
- API-based data collection and processing
- Data quality verification and cleansing processes
- Real-time data updates and synchronization

System Definition

System Architecture

The Productivity Monitoring System is configured based on a 3-tier architecture, consisting of presentation layer (dashboard), business logic layer (data processing and analysis), and data layer (DATA mart integration).

System Components

- Data Collection Module: Real-time data collection through DATA mart integration via Cursor AI
- Data Processing Engine: Cleansing, transformation, and calculation processing of collected data
- Analysis Engine: UPH calculation, achievement rate analysis, trend analysis
- Visualization Module: Various forms of data visualization including charts, graphs, and tables
- Notification System: Alerts for target underachievement or anomaly situations

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System Operation Method

- 24/7 real-time monitoring operation

- Access and inquiry through web-based dashboard
- Role-based access control management
- Data backup and recovery system operation

Dashboard Configuration

- KPI Indicators
- Filter Options
- Graph Areas
- Target UPH vs Actual UPH by Line
- UPH Achievement Rate by Line

Detailed Data Table

Production Date	Line Name	Product Group	Production Qty	Defect Qty	Work Hours	Target UPH	Actual UPH	Company
2024-01-15	Line A	Electronics	1,904	24	8.0	245	238	Samsung Electronics
2024-01-15	Line B	Semiconductor	1,800	18	8.0	220	225	SK Hynix
2024-01-15	Line C	Display	1,560	32	8.0	200	195	LG Display
2024-01-15	Line D	Battery	1,680	15	8.0	210	210	LG Energy Solution
2024-01-15	Line E	Auto Parts	1,440	28	8.0	185	180	Hyundai Mobis
2024-01-15	Line F	Medical Device	1,200	12	8.0	150	150	Samsung Biologics

Expected Effects

Quantitative Effects

- 5-10% average productivity improvement through productivity monitoring
- 20% reduction in downtime through real-time anomaly detection
- 15% improvement in defect rates through data-driven decision making
- Improvement in line-specific achievement rates through target management system establishment

Qualitative Effects

- Swift problem resolution through real-time visibility
- Establishment of data-driven decision-making culture
- Systematization of continuous improvement activities
- Enhanced ease of production status understanding for management